Japanese Anime Styled Vegetation Asset Documentation

Stylized Vegetation for Japanese-Themed Environments

Version: 1.0
Author: AZSoftStudio

Overview

Asset Name:

Majestic Japan: Japanese Anime Styled Vegetation

Purpose

This asset provides stylized vegetation designed for creating Japanese-themed environments. It includes essential features for both aesthetic appeal and performance optimization, making it ideal for game developers, environment designers, and artists.

Key Features:

- **Level of Detail:** Ensures optimized performance for different platforms, including PC, Mobile, and VR.
- **Subsurface Scattering Shader:** Achieve realistic lighting effects for more immersive vegetation.
- **Customizable Shader:** Easily adapt the look to fit various project themes.
- **Aero Flow Wind System**: Adds an anime-stylized wind effect that simulates the movement of trees and foliage, including tree bending and leaf movement based on wind direction, speed, and force.
- **Editor Script:** Allows easily create wind system and control of the wind system parameters, providing global control for tree behavior.

Target Audience

- Game developers seeking ready-to-use assets for Japanese-themed projects.
- Artists and designers looking for customizable vegetation.
- Developers prioritizing performance optimization in stylized environments.

Installation

Unity Version Requirements

- The asset uses API that changed in **Unity 6000.0.xxf1 LTS** and if used in older versions of unity you may need to change those API accordingly.
- Ensure the **Built-in** and **URP** or **HDRP** pipelines are installed, depending on your project requirements.

Importing the Asset

1. Open your Unity project.
2. Go to the **Asset Store** and download the " Majestic Japan: Japanese Anime Styled Vegetation"
3. Once downloaded, click import to import the package into Unity.
4. In the Import dialog, ensure all necessary files are selected, then click **Import**.

Folder Structure

After importing, your project will include:

- **Assets/AZSoftStudio/ Majestic Japan: Japanese Anime Styled Vegetation/**
    - **Vegetation/**: Pre-made vegetation prefabs (Cherry Blossom, Maple, etc.).
    - **Shaders/**: Custom shaders, including the Aero Flow Wind Shader.
    - **Textures/**: Stylized textures for all vegetation.
    - **AeroFlowWind/**: Includes the lite version of Aero Flow Wind System.
    - **Scripts/**: Includes the Aero Flow Wind System script and editor tools.
    - **Showcase/**: Contains the example scene for demonstration purposes.

Quick Start Guide

Setting Up Vegetation in Your Scene

1. **Add Vegetation Prefabs**
    - Navigate to **Assets/AZSoftStudio/ Majestic Japan: Japanese Anime Styled Vegetation/ {Select Your Favorite} /Prefab/**.
    - Drag and drop the desired tree prefab (e.g., Cherry Blossom or Bamboo) into your scene.
    - Adjust the position and scale as needed.
2. **Apply the Aero Flow Wind System**
    - In the Unity toolbar, go to **Tools > AZSoftStudio > Create Aero Flow Wind System**.
    - This will create a GameObject with the Aero Flow Wind System script attached.
    - Adjust the following parameters in the script:
        - **Wind Direction**: Set the wind direction as a vector (e.g., (1, 0) for wind from the x-axis).
        - **Bend Value**: Adjust the float value to control tree bending intensity.
        - **Wind Speed**: Modify the float to control the wind speed effect.
3. **Preview the Wind Effect**
    - Works in runtime, see the trees respond to the wind system in editor.
    - Make real-time adjustments to the parameters in the Inspector to fine-tune the effect.

Adding Textures to Your Own Models

1. Open the **Textures/** folder in the asset directory.
2. Assign the stylized textures to your custom models using Unity’s Material Inspector.
3. Apply the wind shader to enable the Aero Flow Wind System.

Detailed Features - Vegetation Assets

Stylized Vegetation Overview

This asset includes five unique tree types, each designed to enhance the aesthetic of Japanese-themed environments.

1. **Cherry Blossom Tree (Sakura)**
   - Features vibrant pink blossoms for a serene and iconic look.
   - Suitable for spring-themed settings or tranquil garden scenes.
2. **Maple Tree**
   - Displays rich red and orange hues, ideal for autumn-themed environments.
   - Adds warmth and depth to landscapes.
3. **Bamboo Tree**
   - Includes dense clusters for creating secluded or natural pathways.
   - Works well in zen gardens and forested areas.
4. **Pine Tree (Matsu)**
   - Characterized by its unique, stylized shape and evergreen foliage.
   - Perfect for traditional Japanese landscapes and mountainous regions.
5. **Ginkgo Tree**
   - Exhibits golden leaves, adding a touch of elegance to fall-themed environments.
   - Provides diversity in foliage color and texture.

Level of Detail (LOD) System

- **Five LODs**: Each tree includes five levels of detail for optimized rendering.
  - **LOD 0**: High-quality model for close views.
  - **LOD 1-3**: Gradually reduced details for medium to far distances.
  - **LOD 4 (Impostor)**: Simplified 2D billboard for maximum performance.
- **Cross-LOD Animation**: Smooth transitions between LODs, reducing pop-in effects.

Customizability

- All tree prefabs are customizable in terms of scale, material, and positioning.
- Stylized textures can be swapped or edited for unique variations.

Detailed Features - Aero Flow Wind System

Overview

The Aero Flow Wind System is a lightweight yet powerful solution for creating stylized wind effects. It provides global control over the wind's behavior, allowing seamless interaction across all vegetation assets in the scene.

Key Features

1. **Wind Direction**
   - Defines the direction of the wind as a vector (e.g., (1, 0) for wind blowing along the x-axis).
   - Affects both leaves and tree bark, simulating natural wind dynamics.
2. **Bend Value**
   - A float parameter controlling how much the trees bend in response to the wind.
   - Adjust this value for dramatic or subtle effects based on the scene's requirements.
3. **Wind Speed**
   - Modifies the intensity and speed of the wind effect, impacting how quickly the trees sway.
4. **Dynamic Wind**
   - It's a boolean that enables wind dynamics (Random winds based on Objects Position)
5. **Randomness**
   - A float to control the intensity of wind dynamics.

System Setup

1. **Editor Script**
   - Access the system via **Tools > AZSoftStudio > Create Aero Flow Wind System** in the Unity toolbar.
   - Automatically creates a GameObject with the Aero Flow Wind System script attached.
   - Adjusting the parameters on the Aero Flow Wind System GameObject updates all vegetation in real-time.

Future Upgrades

- Plans for additional controls, such as Wind zone painter turbulence and gust effects.
- Enhanced compatibility with custom shaders and other environmental assets.

Detailed Features – Shaders

Stylized Textures

The asset includes handcrafted textures designed to give a stylized, anime-inspired aesthetic to the vegetation.

1. **Oil Paint Effect**
   - All textures were processed with an oil paint filter in Photoshop for a cohesive, artistic style.
   - Enhances the visual appeal, particularly in cartoon-like or anime-themed environments.
2. **High Resolution**
   - Textures are optimized to provide a balance between quality and performance. { 2k resolution can be reduced}
   - Suitable for close-up shots and distant views.
3. **Customizable Colors**
   - Easily adjust the texture colors in Photoshop or Unity's material editor to fit different environmental themes.
   - Seasonal variations (e.g., spring, summer, autumn) can be created with minor adjustments.

Material Setup

This package include a custom shader that have following features and properties:

- **SSS Color:** Controls the tint of scattered light.
- **SSS Intensity**: Adjusts the strength of the scattering effect.
- **SSS Power:** Defines the softness or sharpness of light diffusion.
- **Normal Influence:** Alters the scattering based on surface normals**.**

Usage Tips

- For enhanced performance, reduce texture resolution in inspector for distant objects or less critical assets.
- Consider adding additional post-processing effects (e.g., bloom, vignette) for a polished, stylized look.

Compatibility

This asset is designed for Unity versions **2020.3 LTS and later**. Please note that due to API changes in **Unity 6**,

- **Object.FindObjectOfType()** has replaced **Object.FindFirstObjectByType()** in "***AeroFlowWindShaderEditor***".
- If you are using Unity versions prior to 2023.1, you should continue using **Object.FindObjectOfType()** instead.

How to update API for older Unity Versions (x<6)

You can change the following statement with this :

var existingObject = Object.FindFirstObjectByType<AZSoftStudio.AeroFlow.AeroFlowWindShader>();

Or,

You can find the full code in our Website:

Copy Editor Full Code

- For any assistance in making these adjustments, feel free to Contact Us

Optimization Guidelines

Optimizing Vegetation for Performance

1. **Use of LODs**
    - Ensure that all vegetation assets in your scene utilize the built-in LOD system.
    - **LOD 0** (High-quality): Use sparingly for close-up shots.
    - **LOD 1-4** (Optimized): Cover medium to far distances effectively.
    - **Impostor LOD (LOD 4)**: Ideal for distant objects to save performance.

2. **Batching and Instancing**
    - Enable GPU instancing for vegetation materials to reduce draw calls.
    - Group similar tree types to take advantage of Unity's static batching.

3. **Texture Optimization**
    - Use lower resolution textures for distant assets or less critical objects.
    - Compress textures using Unity's texture compression formats (e.g., ASTC for mobile).

Wind System Efficiency

- The Aero Flow Wind System uses global shader values to minimize overhead.
- Avoid creating multiple wind system GameObjects; one system controls the entire scene and creating more than one system may cause the wind to not work.{Look for more in Troubleshooting Common Issues at 5}
- Keep **Bend Value** and **Wind Speed** moderate to maintain a balance between visual effects and performance.

Additional Recommendations

- Use Unity's **Profiler** to monitor frame rates and identify bottlenecks.
- Enable **Occlusion Culling** in large scenes to reduce rendering of unseen objects.
- Adjust **Quality Settings** in Unity to match the target platform's performance capabilities.

Future Optimization Plans

- A fully optimized Japanese Garden environment is in development and will be included in future updates.
- Enhanced vegetation shaders for even better performance on various platforms.

Support and Contact Information

Support

If you encounter any issues or have questions about the Majestic Japan: Japanese Anime Styled Vegetation, we're here to help.

1. **Documentation**
   - Refer to this documentation for setup, usage, and optimization tips.
   - Check the FAQ section for quick answers to common issues (see Page 11).
2. **Community Support**
   - Visit our online Discord Server mentioned below to interact with other users and share tips.
3. **Bug Reports and Feature Requests**
   - If you find a bug or have a feature suggestion, please let us know.
   - Submit bug reports or requests in our website or in our discord mentioned below.

Contact Information

- **Developer**: AZSoftStudio
- **Website**: azsoftstudio
- **Contact Us:** azsoftstudio/support
- **Social Media**: Follow us on:
  - Twitter: @AZSoftStudio
  - Youtube: @AZSoftStudio

Office Hours

- Friday and Saturday: 7:00 AM - 9:00 PM (GMT+5:30)
- Response Time: Within 24-48 hours for all inquiries.

FAQ and Troubleshooting

Frequently Asked Questions (FAQ)

1. **What Unity versions are supported?**
    - The asset is tested and supported on Unity 2020.3 LTS and later versions.
2. **Can I use this asset for mobile platforms?**
    - Yes, the asset is optimized for mobile use, provided you enable LODs and optimize textures appropriately.
3. **Is the example scene included in the asset package?**
    - The example scene is included for demonstration purposes but is not optimized for deployment.
4. **Can I customize the textures?**
    - Absolutely! You can modify textures to suit your project’s theme while maintaining their stylized aesthetic.
5. **Does this asset support HDRP/URP?**
    - The asset is designed for the Built-in Render Pipeline, HDRP and URP.

Troubleshooting Common Issues

1. **Vegetation does not appear in the scene:**
    - Ensure the values in aero flow wind system are correct and the prefabs are placed in the correct layer and are not hidden by terrain or other objects.
2. **Performance is lower than expected:**
    - Verify that LODs are enabled.
    - Enable GPU instancing for vegetation materials.
    - Use Unity’s Profiler to identify bottlenecks.
3. **Textures look blurry or pixelated:**
    - Check the import settings for the texture and adjust the resolution or compression format.
4. **Shadows are not rendering properly:**
    - Verify your lighting and shadow settings in the Quality settings.
5. **Wind effects are not visible:**
    - Ensure the correct global parameters (1,1|2|1) are being applied in your shaders.
    - Make sure that only one wind system is active on the scene. If there is multiple make sure to delete them or disable them.

End Notes

Final Words

Thank you for choosing the Japanese Garden Vegetation asset for your project. This asset is designed to enhance your environments with stunning stylized vegetation while maintaining optimal performance across platforms.

We hope this documentation has provided all the guidance you need to effectively use the asset. Should you require further assistance, please do not hesitate to reach out to our support team.

Future Updates

We are constantly working to improve this asset. Here's what you can expect in future updates:

1. Fully optimized and included Japanese Garden environment.
2. Additional vegetation types and seasonal variations.
3. Enhanced shaders and performance tweaks for HDRP and URP support.

Acknowledgments

Developed by **AZSoftStudio**. Special thanks to our community for their continued support and feedback, which helps us improve with every update.

This concludes the documentation. If you have suggestions for improvement, let us know.
Happy creating!

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